Barnebys Pilot Analysis - Olséns Auctioner

Technical Documentation

26th February 2026

1. Executive Summary

This document describes the technical implementation of the Barnebys pilot analysis for Olséns Auctioner (auction_house_Id: 3687), covering auction data for 2025 (excluding June–July).

Objective

Primary Objective: Measure the incremental value Barnebys users bring to Olséns auctioneer by analyzing bidding behavior compared to users from other sources.

Secondary Objectives:

1. Establish a replicable data processing pipeline for auction house analysis 2. Identify data quality issues and develop solutions 3. Create a comprehensive metrics framework for auction performance 4. Provide actionable insights on user behavior differences

Technical approach

A three-layer BigQuery pipeline (Raw → Processed → Analytics) links bid events across four systems: Bite, Skeleton, Barnebys AWS and Azure, and auction house data.

Scope

Olséns pilot only. Pipeline designed for future expansion to all Skeleton clients.

2. Business Context

Background

Barnebys is a global auction aggregator platform that drives traffic to auction houses.

Skeleton is auction management software used by 46 auction houses. Each client runs their own website (e.g. Olséns) where users can register, bid, and win auctions.

Bite is Barnebys' tracking and analytics system. Each Skeleton client website embeds a Bite tracking code to measure user behavior and attribute actions back to Barnebys traffic.

This pilot project with **Olséns Auctioner** aims to quantify the value that Barnebys users provide, measured through:

- User engagement (clicks, registrations, bids)

- Conversion (bidders to winners)

- Revenue impact (winning bids and commissions)

- Price formation and incremental value

- Category distribution

Analysis scope

Time Period: December 2024 - December 2025

Analysis Period: Excludes June-July 2025 (incomplete source data)

Data Sources and extraction period:

- Bite clicks and registrations (events), Dec 2024 – Dec 2025
- Bite bids (events), Dec 2024 – Dec 2025
- Barnebys AWS lots, Dec 2024 - May 2025
- Barnebys AZURE lots, Jun 2025 - Dec 2025
- Skeleton Olsens DB, Jan 2025 - Dec 2025

3. Data Architecture

Three-Layer Architecture

All data processing is implemented in BigQuery, following a three-layer pattern:

Layer	Purpose	Tables
Raw	Extract and upload raw data to BigQuery	bite_raw_registrations bite_raw_bids aws_barnebys_raw_lots azure_barnebys_raw_lots skeleton_raw_auctions skeleton_raw_enteredbids raw_auction_house
Processed	Clean, match, and enrich records	proc_skeleton_auctions_with_bite_bids, proc_skeleton_auctions_with_enteredbids, proc_skeleton_auctions_with_winning
Analytics	Aggregated metrics for analysis	ana_funnel_olsens_exclued67 ana_lot_price_tiers ana_bids_with_price_tier ana_winning_bids_with_price_tier ana_barnebys_increment

Data Sources

Source	Database	Key tables	Key fields
Bite	Barnebys-analytics	tracking-clicks tracking-events	Programid, clicks, registrations, bids, sessionid, source, timestamp
Skeleton	Olsens	Auction Inventory Inventorywon Enteredbid Bidauditlog winnertracking	Auction, enddate, inventoryid, auctionsessionid, enteredbidid, webuserid, winnerid, amount, price
Barnebys AWS	barnebys	lots_archived_31052025 categories	Url, auction_house_id, category_name, created
Barnebys Azure	barnebys	lots categories	Url, auction_house_id, category_name,

			created
Generated		Auction_house_list	Programid, auction_house_id,, is_skeleton_client

Data Flow

```
Bite (clicks, bids)      Skeleton (enteredbids)      AWS + Azure (lots)      Auction_house
       ↓                          ↓                          ↓                      ↓
 bite_raw_clicks          skeleton_raw_auctions       raw_lots (merged)      raw_auction_house
 bite_raw_bids            skeleton_raw_enteredbids
 bite_raw_registrations
       ↓                          ↓                          ↓                      ↓
  |_____________________|___________________|_____________________|
                                  ↓
                   proc_skeleton_auctions_with_bite_bids
                                  ↓
                   proc_skeleton_auctions_with_enteredbids
                                  ↓
                   proc_skeleton_auctions_with_winning
                                  ↓
                      ana_funnel / ana_increment
                      ana_price_tiers / ana_bids_tier/ ana_winning_tier
```

Key tables

Input tables:

Table	Description	Record Count
bite_raw_clicks	Raw click events from Bite	61,937
bite_raw_registrations	User registration events	2,724
bite_raw_bids	Raw bid events from Bite	76,400
skeleton_raw_auctions	Auction & inventoryId records	18,553
skeleton_raw_enteredbids	All entered bid records	72,209
aws_barnebys_raw_lots	Lot data from AWS	9,551

azure_barnebys_raw_lots	Lot data from Azure	9,482

Output tables:

Table	Description	Record Count
proc_skeleton_auctions_with_enteredbids	Bids matched to EnteredBid with WebUserid & source	67,286
proc_skeleton_auctions_with_winning	Bids with winning flags & commissions	53,226
ana_funnel_olsens_exclued67	Funnel metrics by user source	64 rows
ana_barnebys_increment	Lot-level Barnebys price increment	10,289

4. Data Processing Logic

Lot Deduplication

AWS and Azure lot sources are merged. For duplicate **inventoryId** records across sources, only the record with the latest created timestamp is retained.

Bite Bids Deduplication

For the same **inventoryId** + **value** combination, only the earliest bid (first timestamp) is kept to avoid double-counting repeated bid events.

Skeleton Auction - Bite Bids Join

Bite bids are joined to Skeleton auction/inventory records using **inventoryId** as the primary key. The scope of Skeleton records is defined by **auction enddate** (2025-01-01 to 2025-12-31), aligned with Skeleton's logic. This means:

- Bids placed in Dec 2024 may be included if the auction ended in Jan 2025
- Bids placed in Dec 2025 are excluded if the auction ended in Jan 2026

The Bite extraction period is therefore extended to **Dec 2024 – Dec 2025** to capture cross-boundary auctions.

Records where both Bite data and hammeredprice are absent are dropped.

EnteredBid Matching

Bite bids are matched to Skeleton's EnteredBid records to retrieve **WebUserid** and **EnteredBidId**. Two strategies applied in order:

Strategy	Matching Criteria	Coverage
Strict	inventoryId + amount + timestamp difference ≤ 1s	99.2%
Relaxed	inventoryId + amount only, closest timestamp	0.6%
Unmatched	No match found	<0.1%

Where the same **inventoryId + amount + timestamp** matches multiple EnteredBid records, the record with the lowest EnteredBidId is retained.

Winning Record Handling

Records with **url IS NULL** represent winning bids recorded directly in Skeleton but not captured in Bite (88 lots). These are handled by direct field mapping rather than EnteredBid matching:

- EnteredBidId ← bidid
- WebUserid ← winnerid
- value ← hammeredprice
- source, sessionId, timestamp ← NULL
- match_type = winning

Source Inference

For records where **source** is still **NULL**, source is inferred from the user's full bid history across all lots:

- Count barnebys vs other bids for that **WebUserid**
- Majority source wins
- If tied → use source from the user's earliest bid

Winning Sign Assignment

Three-rule hierarchy applied in order per inventoryid:

Rule	Condition	Result
Rule 1 (Explicit)	enteredbidId = bidid	win
Rule 2 (Inferred)	No bidid match + value ≥ hammeredprice	win
Rule 3 (Synthetic)	No win found + max(value) < hammeredprice	Create synthetic record with value = hammeredprice

All win **values** are subsequently replaced with hammeredprice to ensure consistency.

User Source Classification

user_source_byfirstbid: Based on the user's first-ever bid on each lot. Once classified as Barnebys, always Barnebys, classification does not change even if subsequent bids come from other sources.

Price Tier Classification

Each lot is assigned a tier based on the distribution of **max_bid_price** across all lots:

Tier	Percentile Range
Low	Bottom 50%
Mid	50% – 80%
High	Top 20%

5. Metrics Definition

Funnel Metrics

User journey from click to winning, classified by source / user_source_byfirstbid :

Metric	Definition	Source Table
clicks	All clicks record from Barnebys	bite_raw_clicks
registrations	Sessionid represent who registered on Olséns from Barnebys	bite_raw_registrations

bidders	Unique WebUserid with at least one bid	proc_skeleton_auctions_with_enteredbids
winners	Unique WebUserid with at least one winning bid	proc_skeleton_auctions_with_winning

Bidding Behaviour Metrics

Metric	Definition	Source Table
avg_bids_per_user	Total bids / unique bidders	proc_skeleton_auctions_with_enteredbids
avg_winning_lots_per_user	Total winning lots / unique winners	proc_skeleton_auctions_with_winning

Revenue Metrics

Metric	**Definition**	**Formula**
total_winning_value	Sum of hammeredprice for all winning lots	SUM(hammeredprice) WHERE winning_sign = 'win'
total_commission	Combined buyerspremium and sellerscommission per winning lot	buyerspremium + currentcommission
ah_commission	Olsens pay to Skeleton at 1.5% commission rate	hammeredprice × 1.5%
revenue_per_click	Commission revenue generated per Barnebys click	ah_commission / clicks Incremental_commission / clicks

Barnebys Incremental Value

Measures the price uplift attributable to Barnebys users through competitive bidding.

Definition: For each lot, the incremental value is the difference between the last Barnebys bid and the highest competing bid immediately before the first Barnebys bid entered the auction.

Calculation logic:

Case	Initial Price	Barnebys Final Bid	Increment
Barnebys placed the first bid	initialbidprice	Highest Barnebys bid ≤ hammeredprice	barnebys_final_bid − initialbidprice
Non-Barnebys bid came first	Highest non-Barnebys bid before first Barnebys bid	Highest Barnebys bid ≤ hammeredprice	barnebys_final_bid − max_bid_before_barnebys

Aggregate metrics:

Metric	Formula
barnebys_increment	SUM(barnebys_final_bid − initial_price_before_first_barnebys_bid)
attributable_commission	barnebys_increment × 0.015
non_attributable_commission	(total_winning_value − barnebys_increment) × 0.015

Price Tier Distribution

Lots are segmented into three tiers based on max_bid_price percentile distribution. Metrics are calculated separately per tier and per user_source:

Metric	Definition
bidders_by_tier	Unique WebUserid with at least one bid in each price tier
winners_by_tier	Unique WebUserid with at least one win in each price tier
winning_value_by_tier	Sum of hammeredprice per price tier
barnebys_share_by_tier	% of winning value attributable to Barnebys users per tier

6. Known Issues & Limitations

InventoryId Extraction Consistency

Olséns has a clean **URL** structure, making **inventoryId** extraction straightforward. However, other auction houses have varying URL formats that may require custom parsing logic, with risk of extraction failures or incorrect mapping.

SessionId as User Identifier

A single user can generate multiple **sessionIds** across sessions (browser changes, cookie expiry, device switching). The fingerprinting cycle resets every 30 days, causing an undercount of returning users and overstatement of unique visitors.

Mitigation: Webuserid from Skeleton matching is used as the primary user identifier where available.

Incomplete Bite Coverage

Bite tracking does not capture all bid records:

- 88 winning lots had no Bite records ➔ handled via direct Skeleton winning records matching
- 136 winning lots had no hammer records ➔ synthetic record creation.

Source Inference Reliability

Users with no directly attributed source may be misclassified if they switch between Barnebys and non-Barnebys sessions, or have very few historical bids.

Skeleton Settlement Timing

Skeleton settles commission based on auction enddate, not individual bid timestamps. Auctions spanning month/year boundaries.

7. SQL Scripts Reference

Script Inventory

Script	Layer	Description
bite_raw_clicks.sql	Raw	Extract clicks records from Bite
bite_raw_registrations.sql	Raw	Extract registration events from Bite

bite_raw_bids.sql	Raw	Extract bid events from Bite
skeleton_raw_auctions.sql	Raw	Extract auction & inventoryId records & hammeredprice from Skeleton
skeleton_raw_enteredbids.sql	Raw	Extract enteredbid records from Skeleton
aws_barnebys_raw_lots.sql	Raw	Extract lot data from AWS
azure_barnebys_raw_lots.sql	Raw	Extract lot data from Azure
raw_bite_bids_clean.sql	Processed	Deduplicate Bite bids
proc_skeleton_auctions_with_bite_bids.sql	Processed	Match Bite bids to Skeleton auctions
proc_skeleton_auctions_with_enteredbids.sql	Processed	Match to EnteredBid, infer source (All bids records)
proc_skeleton_auctions_with_winning.sql	Processed	Assign winning flags and commissions (All winning bids records)
ana_funnel_olsens_exclued67.sql	Analytics	Funnel metrics by user source
ana_lot_price_tiers.sql	Analytics	Price tier classification per lot
ana_bids_with_price_tier.sql	Analytics	All bids enriched with price tier
ana_winning_bids_with_price_tier.sql	Analytics	All winning bids enriched with price tier
ana_barnebys_increment.sql	Analytics	Lot-level Barnebys price increment

8. Dashboard

Overview

The analysis results are visualized in a Looker Studio dashboard, providing an interactive view of Barnebys user performance at Olséns for 2025.

Access: https://lookerstudio.google.com/reporting/5d20a2f5-d762-43d1-bb2f-94e366ff3c16

Data excludes June and July 2025 due to source tracking issues.

Dashboard Pages

Page	Title	Description
1	Barnebys Users Performance	Funnel overview, engagement metrics, winning values
2	Bidding Behavior – Price Formation & Incremental Value	Price tier analysis, Barnebys incremental value
3	Revenue Impact & User Economic Value	Commission attribution, revenue per click/user
4	Barnebys ↔ Olséns Category Segment	Bids share, winning value, revenue concentration by category (Barnebys category)
5	Skeleton ↔ Olséns Category Segment	Same analysis using Olsens' category

How to Refresh

- Re-run SQL script
- Open Looker Studio dashboard
- Click Refresh data on the data source connected to barnebys-skeleton.pilot_olsens